Determinant: Formula Sheet

1. Expansion

Order 2: $ad - bc$

Order 3:

$$a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Cofactor: $C_{ij} = (-1)^{i+j}M_{ij}$

Own row: $\sum a_{1j}C_{1j} = \Delta$

Other row: $\sum a_{1j}C_{2j} = 0$

Cofactor determinant:

$$|C_{ij}| = \Delta^{n-1}$$

2. Properties

Transpose: $D' = D$

Swap two rows: $D \to -D$

Scalar: $|kA| = k^n|A|$

One row $\times k$: kD

$R_i \to R_i + \lambda R_j$: D unchanged

$R_i \to xR_i + yR_j + zR_k$: $D \to xD$

Factor: $f(a) = 0 \Rightarrow (x - a) \mid D$

3. Standard determinants

Cyclic:

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc)$$

Vandermonde:

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a - b)(b - c)(c - a)$$

With a^3:

$$\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)$$

With a^2, a^3:

$$\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(ab + bc + ca)$$

Diagonal 1:

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc + ab + bc + ca$$

Product row:

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ca & ab \end{vmatrix} = (a - b)(b - c)(c - a)$$

4. Multiplication

Product: $|A||B| = |AB|$ (row×column, etc.)

Power sums:

$$|S_{i+j-2}| = [(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)]^2$$

Squares:

$$\left|(a_i + b_j)^2\right| = -2(a_1 - a_2)(a_2 - a_3)(a_3 - a_1)(b_1 - b_2)(b_2 - b_3)(b_3 - b_1)$$

5. Systems: two variables

Unique:

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

Infinite:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

None:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Three lines concurrent:

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

6. Cramer's rule

Solution:

$$x = \frac{D_x}{D},\ y = \frac{D_y}{D},\ z = \frac{D_z}{D}$$

Unique: $D \neq 0$

None: $D = 0$, some $D_x, D_y, D_z \neq 0$

Check case: $D = D_x = D_y = D_z = 0$

Consistency when $D = 0$:

$$R_3 = \lambda R_1 + \mu R_2 \Rightarrow d_3 = \lambda d_1 + \mu d_2$$

Homogeneous: non-trivial $\Leftrightarrow D = 0$

7. Function and equations

Constant term $= \Delta(0)$

Sum of coefficients $= \Delta(1)$

Range of $a + b\sin 2x$: $[a - |b|,\ a + |b|]$

Sum inside: add up only the row (column) that contains r